



APPLICATION FOR OBSERVING TIME

115.2841

IMPORTANT NOTICE

By submitting this proposal, the PI takes full responsibility for the content of the proposal, in particular with regard to the names of Cols and the agreement to act according to the ESO policy and regulations, should observing time be granted.

An ATLAS view on the occurrence of core-collapse supernovae as a function of metallicity

ABSTRACT

Recently, observations have suggested that the occurrence of core-collapse supernovae (CCSNe) is strongly dependent on environment metallicity, with more CCSNe exploding at low metallicity environments. Given the importance of this result to many areas of astrophysics (as it affects our understanding of CCSN explosions, star-formation across galaxies and how the initial mass function (IMF) changes with environment properties), verification is required with distinct SN samples. We request MUSE observations of 99 host galaxies of ATLAS discovered CCSNe to robustly check the possible metallicity effect on the occurrence of these events. The analysis of a new sample will provide results independent of previous survey biases and substantially increase the statistical significance. Additionally, these observations will directly test model predictions of dust evolution, IMF, and delay time distributions of CCSNe across the Universe.

SCIENTIFIC KEYWORDS

stars: supernovae, galaxies: abundances, stars: evolution

RUNS

Run	Period	Instrument	Tel. Setup	Constraints	Mode	Type	Propr. Time	Req. Time
115.2841.001 • Run 1	115	MUSE	UT4	FLI: 100% • Turb.: 85% • pwv: 30.0mm • Sky: Variable, thin cirrus	SM	Normal	12m	99h00m

AWARDED AND FUTURE TIME REQUESTS

Time already awarded to this project

Instrument	Time	Comment
MUSE	99h	113.26FT. SM. Current semester, 73% completed, data to be analyzed.
MUSE	60h	111.24UM. SM. 78% completed, data to be analyzed.
MUSE	99h	096.D-0296(A). SM. 72% completed. All data are reduced and analysed. Published and in prep. papers.
MUSE	99h	103.D-0440(A). SM. 82% completed. All data are reduced and analysed. Published and in prep. papers.

Future time requests to complete this project

- none -

Special Remarks

This project is a 'filler' program, with targets across the full RA and DEC range, that are observable in THN, bad seeing conditions, and during bright time. We understand that only part of the observations may be completed.

This proposal is a continuation of the programme 113.26FT, which was not completed during P113 (73% of the targets observed). We intend to increase the sample size and at least double the number of observed galaxies to obtain statistically significant results.

DESCRIPTION OF THE PROPOSED PROGRAMME

A- Scientific Rationale

Core-collapse supernovae (CCSNe) are luminous explosions associated with the end of the life of massive ($\geq 8 M_{\odot}$) stars. These events are responsible for dramatic changes in the evolution galaxies, triggering or ceasing the formation of new stars. They are also the main formation channel of heavy elements, thus being a fundamental driver of chemical enrichment in galaxies. The study of CCSNe has, therefore, a deep impact not only on the study of stellar evolution but on many fields of astronomy.

Because most massive stars explode as CCSNe, their occurrence is directly linked to the star-formation (SF) rate across the Universe. The rate of CCSNe can therefore be used as a tracer of how SF changes with respect to physical properties, such as age and metallicity. Recently, Pessi+23 used a sample of CCSNe discovered by the All Sky Automated Survey for SuperNovae (ASAS-SN) and showed that **the occurrence of CCSNe has a strong dependence on metallicity**, with these events preferring lower metallicity regions. They show that the difference between the occurrence of CCSN per unit SF between Small Magellanic Cloud (SMC) and solar metallicities has a very large factor of ~ 15 (Fig. 1, left). Additionally, they show that the distribution of CCSN metallicity does not follow the overall SF distribution in their sample. This was an unexpected result, which disagrees with most stellar population and black hole formation models (e.g., Heger+03), and has deep implications not only for massive star evolution, but also other areas of astronomy.

Some of the possible physical reasons that explain this result are: i) **a metallicity effect on the explodability of massive stars** - this would imply that there is a larger number of massive stars exploding as CCSNe at lower metallicity environments, which has strong implications to our current understanding of the mechanism behind CCSN explosions; ii) **selection effects**: a large number of CCSN events might be missed due to intrinsic properties of the survey (to e.g., highly extincted regions) - if this is the case, current CCSN rates are being underestimated by a significant factor; iii) **a metallicity dependence on the initial mass function (IMF)** - the shape of the IMF could be such that it is producing more massive stars per unit SF at low metallicity. This would be strong evidence for a dependence of the IMF on environment properties; iv) **The $H\alpha$ ionization is significantly biased towards massive SF** - this would imply that the $H\alpha$ to star-formation rate (SFR) conversion is significantly affected by metallicity, and that current stellar evolution prescriptions are underestimating this effect by a large factor. Because of the high impact of these results to many fields of astrophysics, affecting our understanding of SF and IMF across different properties of galaxies, **the metallicity dependence on the occurrence of CCSNe should be tested by new samples.**

We propose to observe CCSN host galaxies discovered by the Asteroid Terrestrial-impact Last Alert System (ATLAS) survey (Tonry+18). ATLAS is a recent all-sky high-cadence transient survey dedicated to monitoring small bodies in the solar system. Because it is a panoramic and untargeted survey, it also discovers SNe (and other extragalactic transients) homogeneously. We plan to use the volume-limited ATLAS 100 Mpc sample, which will allow us to test the metallicity dependence on the occurrence of CCSNe with different survey biases than previous studies. Additionally, ATLAS can discover and follow up fainter SNe ($m < 19$ mag) than ASAS-SN ($V < 17$ mag), being more complete in terms of luminosity function, and allowing for a smaller bias towards brighter events. Finally, ATLAS has high-cadence optical transient light-curves (LCs), which allows for a more complete selection and analyses of SN sub-types.

B- Immediate Objective

We propose MUSE observations of **99 ATLAS CCSN host galaxies (76 II, 8 Ic, 6 Ib, 2 IIb, 4 Ib/c, and 3 Ic-BL)**. These targets were randomly selected from all galaxies that hosted CCSNe detected by the ATLAS survey between 2017 September and 2021 March, up to a distance of 100 Mpc, that have a classification spectrum, below a declination of $+30^\circ$, and with no previous MUSE observations. The number of selected galaxies was defined by the time limit of a normal programme (99h), with a total of 1h per target and with all galaxies requiring one pointing. The number of objects selected is also similar to the sample size of previous works (e.g., Pessi+23). MUSE is the only IFU instrument in operation that can provide at the same time exquisite spatial sampling, good wavelength coverage, and a large field of view, allowing for a very detailed characterization of CCSN host galaxies. We will use established techniques of stellar continuum fitting (with STARLIGHT, Cid Fernandes+11), HII regions segmentation (with IFUanalysis, Lyman+18 - Fig. 1, right) and emission line characterization to derive physical parameters (specially lines of $H\alpha$, $H\beta$, O [III], S II and N II, used to estimate age, SFR and metallicity) at a local and global scale within the galaxies. Specifically, our main goal is to:

1) Robustly check the occurrence of CCSNe as a function of metallicity. With a new and more complete (in terms of SN brightness) CCSN sample, we will be able to check the surprising results of Pessi+23 and whether the occurrence of CCSN show a clear dependence on metallicity (i.e., whether the ATLAS sample follows the

same behavior as observed in Fig.1, left). These observations will provide results with distinct survey biases and increase significantly the statistics when combining the samples (e.g., with ASAS-SN). We also plan to estimate the occurrence of CCSNe as a function of other environmental parameters, such as age and extinction. Additionally, we plan to use ATLAS survey rates to compare these results to models of IMF, dust evolution, and delay time distributions of CCSNe across the Universe, which will provide important results for different fields of astrophysics.

Additional results:

These will be the first observations of ATLAS SN host galaxies with MUSE. They will significantly increase the current sample of SN host galaxies observed by the VLT and the data will be available not only for transient but other extragalactic studies. This dataset will additionally allows for a very detailed analysis of CCSN environments and produce significant results. As a product of this sample, we will additionally be able to:

2) Analyze the local environments of ATLAS CCSNe. We plan to analyze the extracted spectrum around the CCSN locations to derive physical parameters of SFR, metallicity, stellar population age, and extinction. A new large sample of CCSNe will allow us to obtain statistically significant results. We will compare the results for the different CCSN subtypes in order to constrain the age and mass ranges of their progenitor stars.

3) Look for correlations between the light curve (LC) properties of CCSNe and their environments. Important physical parameters of the explosion can be derived from the observed LCs of SNe, such as the total ejecta mass, estimated from the LC width, and the ^{56}Ni mass, estimated from the peak absolute magnitude or late-time decline. Previous results show indications for a correlation between these parameters and the physical properties of their local environments for CCSNe (e.g., Galbany+18). Because all ATLAS CCSNe within 100 Mpc have a high signal-to-noise and high-cadence photometry (being more complete than previous surveys, e.g., ASAS-SN), this analysis will provide more significant results than previous works.

Figures

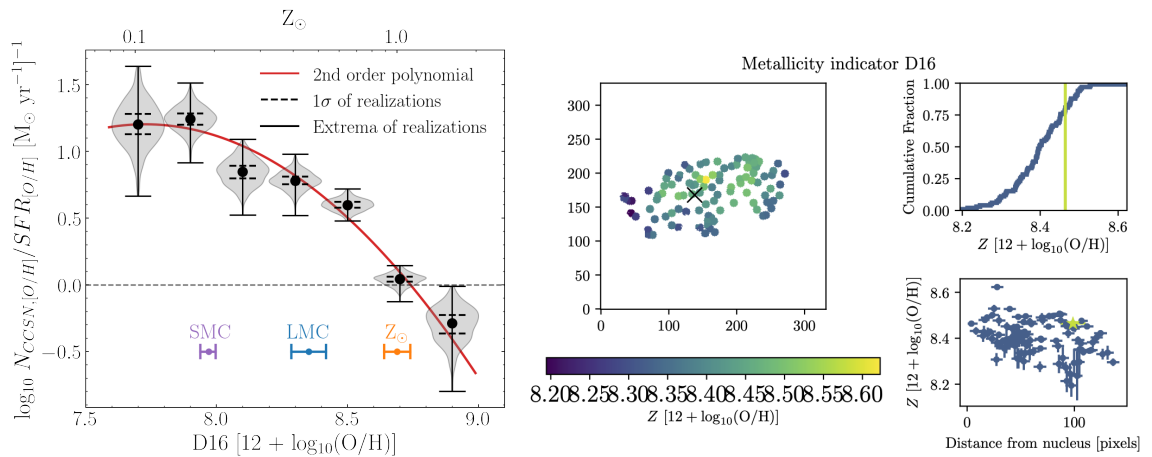


Fig. 1: *Left*: The occurrence of CCSN per unit SF as a function of metallicity derived from the ASAS-SN sample, from Pessi+23. This quantity is estimated by dividing the number of CCSNe by the total SFR over binned intervals in the metallicity axis. The plot shows a strong dependence on metallicity, with CCSNe occurring per unit SF with a much larger factor at low metallicity environments. *Right*: Output from the code IFUanalysis for the MUSE datacube of the galaxy IC4224, that hosted the Type II SN ATLAS17hpc. The left panel shows all the identified HII regions in the galaxy, with the colorbar showing the metallicity given by the D16 index (Dopita+16). The right top and bottom panels show, respectively, the cumulative distribution of metallicity and the metallicity gradient, obtained from these regions, with the SN position marked in green. We plan to use this code to perform HII region segmentation of the observed galaxies, in order to characterize the physical parameters obtained from the spectra of the star-forming regions across those galaxies.

References

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TARGETS

Name	RA	Dec	Coord	Runs	Comment
2020eai	04:59:08.048	+04:58:21.756	J2000	1	SN Ib
V=11.9					
2018jfp	03:17:56.261	-00:10:10.639	J2000	1	SN II
V=11.9					
2020abhs	12:54:57.393	-13:32:29.598	J2000	1	SN II
V=11.9					
2020epi	12:22:08.775	+06:05:59.075	J2000	1	SN Ic
V=11.9					
2020aatb	05:24:46.410	-15:56:31.474	J2000	1	SN II
V=11.9					
2019lre	01:58:50.784	-09:35:05.424	J2000	1	SN II
V=11.9					
2018jlp	11:23:33.235	-08:39:04.777	J2000	1	SN II
V=11.9					
2020xkg	09:14:38.943	-19:33:18.374	J2000	1	SN II
V=11.9					
2020rur	04:59:21.802	+05:37:21.097	J2000	1	SN Ic
V=11.9					
2021os	12:02:54.078	+05:36:53.194	J2000	1	SN II
V=11.9					
2018dfh	05:11:33.458	-21:16:10.268	J2000	1	SN II
V=11.9					
2018ctr	13:48:47.808	+06:21:18.785	J2000	1	SN II
V=11.9					
2018cdc	00:51:05.260	-07:03:47.750	J2000	1	SN II
V=11.9					
2019odp	23:07:19.085	+13:51:21.449	J2000	1	SN Ic-BL
V=11.9					
2020oc	09:48:02.912	+03:32:17.671	J2000	1	SN Ic
V=11.9					
2019fxo	11:03:52.604	-03:20:13.114	J2000	1	SN Ib/c
V=11.9					
2018gvt	23:07:32.542	+23:00:20.898	J2000	1	SN II
V=11.9					
2017iwe	13:33:12.093	-16:07:05.938	J2000	1	SN IIP
V=11.9					
2019xvv	14:24:19.230	-16:22:19.315	J2000	1	SN II
V=11.9					
2020rvf	03:54:15.942	+15:55:42.899	J2000	1	SN Ic
V=11.9					
2021bwq	11:28:17.079	-12:53:17.606	J2000	1	SN Ic
V=11.9					
2018hcp	08:22:57.667	+27:42:10.354	J2000	1	SN IIP
V=11.9					
2019nyk	00:15:15.198	-08:11:21.844	J2000	1	SN II

Name	RA	Dec	Coord	Runs	Comment
V=11.9					
2021gdz	11:53:19.030	+20:06:30.902	J2000	1	SN II
V=11.9					
2019ply	02:59:46.978	+15:17:48.653	J2000	1	SN II
V=11.9					
2020yyz	02:34:00.175	+20:58:26.634	J2000	1	SN IIP
V=11.9					
2019spk	10:02:54.390	-06:12:26.042	J2000	1	SN II
V=11.9					
2019xvo	03:58:22.726	+06:42:55.476	J2000	1	SN II
V=11.9					
2018hwm	08:21:28.182	+03:09:52.427	J2000	1	SN II
V=11.9					
2020aeje	00:17:11.964	-07:37:37.326	J2000	1	SN II
V=11.9					
2020nap	12:07:08.824	+16:55:16.230	J2000	1	SN II
V=11.9					
2018jkb	02:21:09.805	-04:04:23.657	J2000	1	SN II
V=11.9					
2021gno	12:12:10.298	+13:14:57.052	J2000	1	SN II
V=11.9					
2019dke	12:53:38.211	-26:17:31.391	J2000	1	SN II
V=11.9					
2020tsc	08:07:53.306	-18:41:58.362	J2000	1	SN Ic-BL
V=11.9					
2018iqx	22:27:53.594	+19:27:38.484	J2000	1	SN Ib/c
V=11.9					
2017jcu	03:22:46.967	+00:08:53.585	J2000	1	SN II
V=11.9					
2017kam	13:42:43.917	-24:20:06.302	J2000	1	SN II-pec
V=11.9					
2018hi	08:53:40.573	-25:03:31.961	J2000	1	SN II
V=11.9					
2020aaxf	08:20:41.967	-01:24:52.348	J2000	1	SN IIb
V=11.9					
2020uzq	23:34:00.196	-11:46:58.984	J2000	1	SN II
V=11.9					
2020acbm	02:40:17.795	+02:25:37.542	J2000	1	SN II
V=11.9					
2018bjo	12:51:19.856	+21:13:49.040	J2000	1	SN IIP
V=11.9					
2019zc	00:29:29.399	-13:28:42.733	J2000	1	SN II
V=11.9					
2019hgu	00:50:24.894	-12:05:42.673	J2000	1	SN II
V=11.9					
2020zoq	10:36:21.592	+13:43:07.205	J2000	1	SN II
V=11.9					

Name	RA	Dec	Coord	Runs	Comment
2021bjk V=11.9	02:01:52.221	-10:27:48.906	J2000	1	SN Ib/c
2018kfn V=11.9	16:02:42.480	+15:14:11.263	J2000	1	SN II
2020abtf V=11.9	07:59:34.479	+15:18:00.025	J2000	1	SN II
2021iy V=11.9	11:18:31.673	-06:16:40.501	J2000	1	SN II
2018afm V=11.9	07:59:25.015	+16:25:34.619	J2000	1	SN II
2017hca V=11.9	08:49:41.068	-08:05:31.218	J2000	1	SN II
2018hle V=11.9	03:39:28.109	-13:07:02.384	J2000	1	SN II
2019tls V=11.9	08:02:49.927	+15:48:05.537	J2000	1	SN Ib
2019pjs V=11.9	18:04:40.344	+21:38:04.272	J2000	1	SN II
2021aek V=11.9	11:59:48.027	-21:23:12.905	J2000	1	SN II
2020xqj V=11.9	12:03:49.533	+29:43:01.790	J2000	1	SN II
2020vzv V=11.9	21:13:24.762	+17:40:29.374	J2000	1	SN II
2020htt V=11.9	10:31:38.369	-17:51:08.114	J2000	1	SN Ib
2020jjj V=11.9	14:31:19.635	-25:39:31.111	J2000	1	SN II
2020bij V=11.9	10:55:11.698	-26:08:24.518	J2000	1	SN II
2018bui V=11.9	14:20:43.446	+13:41:43.343	J2000	1	SN II
2017ilo V=11.9	03:31:20.633	+04:07:54.584	J2000	1	SN II
2018lcd V=11.9	01:51:10.660	-03:29:24.853	J2000	1	SN II
2018cho V=11.9	00:13:26.503	+17:29:19.543	J2000	1	SN IIP
2020szs V=11.9	21:34:08.994	+08:46:46.366	J2000	1	SN II
2019env V=11.9	09:09:35.638	+29:58:14.534	J2000	1	SN II
2019hnl V=11.9	23:43:10.247	-02:56:58.873	J2000	1	SN II
2020adhh V=11.9	04:43:46.158	+28:58:10.564	J2000	1	SN II
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Name	RA	Dec	Coord	Runs	Comment
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V=11.9					
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2018cqi	13:57:40.915	+17:30:29.142	J2000	1	SN IIP
V=11.9					
2020able	09:26:02.269	+24:31:18.152	J2000	1	SN Ic
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V=11.9					
2018gle	04:25:03.013	-14:36:47.804	J2000	1	SN Ic
V=11.9					
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V=11.9					
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V=11.9					
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V=11.9					
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V=11.9					
2020adow	08:33:42.263	+27:42:43.528	J2000	1	SN Ic-BL
V=11.9					
2019eez	06:16:22.222	-21:21:35.712	J2000	1	SN II
V=11.9					
2020uas	22:59:38.659	+20:08:35.686	J2000	1	SN Ib/c
V=11.9					
2020hox	19:42:26.160	-15:52:44.018	J2000	1	SN II
V=11.9					
2018bif	11:45:44.824	-11:47:17.041	J2000	1	SN II
V=11.9					
2021ejn	10:37:14.439	+04:33:07.682	J2000	1	SN II
V=11.9					
2020abf	09:18:05.927	-15:48:44.708	J2000	1	SN Ic
V=11.9					
2019osl	04:38:54.326	+18:50:10.201	J2000	1	SN II
V=11.9					
2019bzd	14:47:32.040	-19:45:57.629	J2000	1	SN IIP
V=11.9					
2019tba	04:55:09.845	-16:09:01.451	J2000	1	SN II
V=11.9					
2018rz	14:27:11.437	-02:16:11.060	J2000	1	SN IIP
V=11.9					
2020kxg	13:20:00.505	-16:39:40.295	J2000	1	SN Ib
V=11.9					
2020acat	11:50:41.228	-10:13:38.892	J2000	1	SN I Ib
V=11.9					

Name	RA	Dec	Coord	Runs	Comment
2018jsn V=11.9	11:59:57.007	-00:33:01.728	J2000	1	SN II
2019lsj V=11.9	19:36:59.137	-11:57:13.655	J2000	1	SN IIP
2019qba V=11.9	15:37:44.726	+22:25:36.833	J2000	1	SN II
2017gww V=11.9	00:03:49.218	+07:28:00.289	J2000	1	SN II
2020ltq V=11.9	21:44:34.458	-24:36:10.127	J2000	1	SN Ib
2020ekk V=11.9	16:44:45.150	+22:30:47.894	J2000	1	SN II

Target Notes

This dataset contains 99 galaxies randomly selected from a sample of all galaxies that hosted core-collapse supernovae detected by the ATLAS survey between 2017 and 2021, up to a distance of 100 Mpc, below a declination of 30 degrees, and with no previous MUSE observations. The number of selected galaxies was defined by the time limit of a normal programme (99h), with a total of 1h per target and with all galaxies requiring one pointing.

REMARKS & JUSTIFICATIONS

Lunar Phase and Constraints Justification

Please justify here the requested lunar phase and other observing constraints.

Our targets can be observed in any lunar phase as this project is a 'filler' program.

Time Justification

Please describe here a detailed computation of the necessary time to execute the observations, including time-critical aspects if any. Parameters used in the ETC should be mentioned so the computation can be reproduced.

Our proposed MUSE observations aim to detect and measure the strength of H II region emission lines throughout target galaxies, together with narrow absorption features in the galaxy continuum. To estimate exposure times, we assume a typical r-band surface brightness of 21 mag per arcsec² for faint HII regions (James et al. 2004). We then use the MUSE ETC to estimate our required exposure times. We assume an 'Infinitely extended source' and the HII region template (at a redshift of 0.05) together with an r-band surface brightness of 21 mag per arcsec². As a 'filler' proposal we assume observations to 7 days from new moon with an airmass of 1.4 and seeing 1.5".

In order to remove the edges of each IFU on the detector (artifacts of the image slicer), it is best to combine observations with all four 90 degree angles. With 4×700 second exposures, the ETC gives a S/N of ~5 in the continuum, which translates to a S/N of more than 100 for all the emission lines we wish to detect (e.g., H α , [Nii], [O iii], H β). This totals roughly 1 hr per pointing. All the galaxies in our sample are covered by one MUSE pointing, and we thus require 99 h to observe our sample of 99 host galaxies.

We have previously obtained MUSE data with very similar conditions/observing strategy to that outlined above, and these exposure times allowed us to make the necessary measurements. We have already published several papers based on these data (see publications section). Our team has experts on the reduction, analysis and interpretation of IFS galaxy observations, while IFS data and their analysis can sometimes prove somewhat daunting, our team provides all the required knowledge to fully exploit this rich data set. These data will remain in the archive as a legacy for future extragalactic studies.

Telescope Justification

Please justify why the telescope requested is the best choice for this programme.

MUSE at the VLT is the only currently available IFU instrument that has a FoV of sufficient size to cover the majority of our targets, while at the same time providing high spatial resolution, and sensitivity, allowing targets to be observed in a relatively small amount of time.

We stress here that even in relatively bad seeing conditions we will still be able to unravel kinematic and population substructure in these galaxies over a large FoV. Indeed, even in bad seeing conditions these data will still be better able to probe SN environments and galaxy characteristics than any current or past IFU survey (SAURON, ATLAS3D, CALIFA, SAMI, MaNGA).

Observing Mode Justification

Please justify the choice of SM, VM or dVM.

Targets are observable throughout the semester, so service mode is required.

Calibration Request

If you need any special calibration not included in the instrument calibration plan, please specify it here.

Standard Calibration.

Duplication with ESO Science Archive

If observations of the same target(s) using the same instrument(s) already exist in the ESO archive, please justify why this programme requests further observations.

We searched for duplications, and there are no IFU observations of these galaxies in the ESO or any other public archive.

GTO Target Duplication Justification

If an instrument GTO team aims at the same target(s), please justify why this programme requests further observations.

n/a

Background and Expertise

Short description of the background, expertise and roles of the various team members in the context of the science case discussed in the proposal. For small teams the applicants may wish to provide a sentence for the qualifications of each member, while for larger teams (e.g. in Large Programmes), only the leading roles need to be specified. [Non-anonymised]

The team has acquired and analysed the vast majority of VLT/MUSE observations of supernova host galaxies. We have tools in place to rapidly leverage the scientific yield of this combined data set. The AMUSING project has run for 12 previous semesters, and has already published >30 papers (for a complete list of publications check <https://amusing-muse.github.io/publications/>), with many other projects currently ongoing

T. Pessi is a postdoctoral fellow working on supernova observations and has experience in using MUSE to study host galaxies of supernovae. J. Prieto is an expert in supernova observations and has also worked in host galaxy observations/analysis using MUSE. He is part of the ASAS-SN team.

J. P. Anderson has previously led the AMUSING proposals, and thus has significant experience of data and project organisation. He has expertise in supernova environment studies and how those studies can constrain progenitor scenarios.

J. Lyman is an expert in the study of various transient events and the analysis of SN environments with MUSE. He has developed the code IFUanalysis, which was used to analyze MUSE datacubes of several galaxies that hosted SNe and has been used in several publications of the field

L. Galbany is an expert in studying various supernovae and their host galaxies with integral field spectrograph (IFS) data. He has led and published several works from different campaigns on SNe and their host galaxies, and has expertise in the scheduling, reduction and analysis of VLT data.

S. González-Gaitán is a SN Ia cosmology expert and has led several analyses in large SN surveys (SDSS, SNLS, DES).

H. Kuncarayakti has led MUSE core-collapse SN host galaxies projects (AMUSING, FOSSIL), and is an expert on analysing IFU data.

F. Förster is an expert in supernova observations and astroinformatics, having experience with large transient surveys (HiTS, DES). He is currently leading the ALeRCE project.

S. Srivastav, M. Nicholl, T. Moore, and S. Smartt are experts in the discovery and studies of different transient events, and have lead many important projects in the field. They are part of the ATLAS team.

S. F. Sanchez, F. F. Rosales-Ortega and J. Falcon-Barroso are experts in galaxy formation and evolution, particularly through the use of spectral synthesis with IFS data. They have lead and worked on many projects and publications with data obtained from large IFS galaxy surveys (SDSS, CALIFA, PINGS).

T. Kravtsov is a graduate student working on MUSE observations of SN host galaxies.

REPORT ON PREVIOUS USAGE OF ESO FACILITIES

Run	PI	Instrument	Time	Mode	Comment
111.24UM.001	Thallis Pessi	MUSE	60.0h	Service	78% completed, data to be analyzed.
The local environments of stripped-envelope supernovae					
113.26FT.001	Thallis Pessi	MUSE	99.0h	Service	Current semester, 73% completed, data to be analyzed.
An ATLAS view on the occurrence of core-collapse supernovae as a function of metallicity					
109.2399.001	Jose Prieto	MUSE	13.0h	Service	3h completed so far.
The Extreme Host Galaxy Environments of Low-Luminosity Type Ia SNe with late-time H-alpha emission					
109.2399.002	Jose Prieto	MUSE	4.0h	Service	3h completed so far.
The Extreme Host Galaxy Environments of Low-Luminosity Type Ia SNe with late-time H-alpha emission					
0103.D-0440(B)	Joseph Anderson	MUSE	10.0h	Service	Data reduced and analyzed. Has contributed to the >30 published with AMUSING data so far.
The All-weather MUSE Supernova Integral field Nearby Galaxies (AMUSING) survey IX: supernova and tidal disruption event rates as a function of environment age and metallicity					
0103.D-0440(A)	Joseph Anderson	MUSE	89.0h	Service	Data reduced and analyzed. Has contributed to the >30 published with AMUSING data so far.
The All-weather MUSE Supernova Integral field Nearby Galaxies (AMUSING) survey IX: supernova and tidal disruption event rates as a function of environment age and metallicity					
0104.D-0503(A)	Joseph Anderson	MUSE	50.0h	Service	Data reduced and analyzed. Has contributed to the >30 published with AMUSING data so far.

Run	PI	Instrument	Time	Mode	Comment
The All-weather MUse Supernova Integral field Nearby Galaxies (AMUSING) survey X: star formation in low-mass galaxies as traced by extra-galactic transients					
0104.D-0503(B)	Joseph Anderson	MUSE	49.0h	Service	Data reduced and analyzed. Has contributed to the >30 published with AMUSING data so far.
The All-weather MUse Supernova Integral field Nearby Galaxies (AMUSING) survey X: star formation in low-mass galaxies as traced by extra-galactic transients					

RECENT PI/CoIs PUBLICATIONS MOST RELEVANT TO THE SUBJECT OF THIS PROPOSAL

1.

Pessi, T., et al. (2023) "A characterization of ASAS-SN core-collapse supernova environments with VLT+MUSE. I. Sample selection, analysis of local environments, and correlations with light curve properties," A&A, 677, A28 - [2023A&A...677A..28P](#)

2.

Pessi, T., Anderson, J. P., Lyman, J. D., et al. (2023) "A Metallicity Dependence on the Occurrence of Core-collapse Supernovae," ApJL, 955, L29 - [2023ApJ...955L..29P](#)

3.

López-Cobá, C., Sánchez, S. F., Lin, L., et al. (2022) "Exploring Stellar and Ionized Gas Noncircular Motions in Barred Galaxies with MUSE," ApJ, 939, 40 - [2022ApJ...939...40L](#)

4.

Nicholl, M., Srivastav, S., Fulton, M. D., et al. (2023) "AT 2022aedm and a New Class of Luminous, Fast-cooling Transients in Elliptical Galaxies," ApJL, 954, L28 - [2023ApJ...954L..28N](#)

5.

Moriya, T. J., Galbany, L., Jiménez-Palau, C., et al. (2023) "Environmental dependence of Type IIin supernova properties," A&A, 677, A20 - [2023A&A...677A..20M](#)

6.

Cleland, C., McGee, S. L., & Nicholl, M. (2023) "Metallicity beats sSFR: the connection between superluminous supernova host galaxy environments and the importance of metallicity for their production," MNRAS, 524, 3559-3567 - [2023MNRAS.524.3559C](#)

7.

Srivastav, S., Smartt, S. J., Huber, M. E., et al. (2022) "SN 2020kyg and the rates of faint Iax supernovae from ATLAS," MNRAS, 511, 2708-2731 - [2022MNRAS.511.2708S](#)

8.

Robledo-Orús, A. C., Torres-Papaqui, J. P., Longinotti, A. L., et al. (2021) "A MUSE study of NGC 7469: Spatially resolved star-formation and AGN-driven outflows," IAUS, 359, 291-294 - [2021IAUS..359..291R](#)

9.

Srivastav, S., Moore, T., Nicholl, M., et al. (2023) "Unprecedented early flux excess in the hybrid 02es-like type Ia supernova 2022ywc indicates interaction with circumstellar material," arXiv, arXiv:2308.06019 - [2023arXiv230806019S](#)

10.

Grayling, M., Gutiérrez, C. P., Sullivan, M., et al. (2023) "Core-collapse supernovae in the Dark Energy Survey: luminosity functions and host galaxy demographics," MNRAS, 520, 684-701 - [2023MNRAS.520..684G](#)

11.

Uddin, S. A., Burns, C. R., Phillips, M. M., et al. (2023) "Carnegie Supernova Project-I and -II: Measurements of H_0 using Cepheid, TRGB, and SBF Distance Calibration to Type Ia Supernovae," arXiv, arXiv:2308.01875 - [2023arXiv230801875U](#)

INVESTIGATORS

- Thallis Pessi, ESO Chile, ESO (PI)**
- Jose Prieto, Universidad Diego Portales, Chile
- Joseph Anderson, ESO Chile, ESO
- Joseph Lyman, University of Warwick, United Kingdom
- Shubham Srivastav, Queen's University - Belfast, United Kingdom
- Matt Nicholl, Queen's University - Belfast, United Kingdom
- Thomas Moore, Queen's University - Belfast, United Kingdom
- Stephen Smartt, University of Oxford, United Kingdom
- Lluis Galbany, Institute of Space Sciences ICE-CSIC, Spain
- Jesus Falcon-Barroso, Instituto de Astrofísica de Canarias, Spain
- Santiago Gonzalez Gaitan, Universidade de Lisboa, Portugal
- Hanindyo Kuncarayakti, Turun yliopisto, Finland
- Francisco Forster, Universidad de Chile, Chile
- Fernando Fabian Rosales-Ortega, Instituto Nacional de Astrofísica, Óptica y Electrónica, Mexico
- Timo Kravtsov, Turun yliopisto, Finland

OBSERVATIONS

In the table below, the repeat factor is applied to the complete observation on that target, including its overhead.

✓ The PI acknowledged that all the telescope times listed below include overheads.

Run 115.2841.001 • Run 1 • P115 • MUSE • SM

Tel. Time: 99h00m

FLI: 100% • Turb.: 85% • pwv: 30.0mm • Sky: Variable, thin cirrus • Airmass: 2.8

Target • 2020eai • 04:59:08.048 • +04:58:21.756

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

Target • 2018jfp • 03:17:56.261 • -00:10:10.639

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

Target • 2020abhs • 12:54:57.393 • -13:32:29.598

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

Target • 2020epi • 12:22:08.775 • +06:05:59.075

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

Target • 2020aatb • 05:24:46.410 • -15:56:31.474

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

Target • 2019lre • 01:58:50.784 • -09:35:05.424

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

Target • 2018jlp • 11:23:33.235 • -08:39:04.777

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

Target • 2020xkg • 09:14:38.943 • -19:33:18.374

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

Target • 2020rur • 04:59:21.802 • +05:37:21.097

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

Target • 2021os • 12:02:54.078 • +05:36:53.194

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

Target • 2018dfh • 05:11:33.458 • -21:16:10.268

Tel. Time: 01h00m

OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018ctr • 13:48:47.808 • +06:21:18.785		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018cdc • 00:51:05.260 • -07:03:47.750		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019odp • 23:07:19.085 • +13:51:21.449		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020oc • 09:48:02.912 • +03:32:17.671		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019fxo • 11:03:52.604 • -03:20:13.114		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018gvt • 23:07:32.542 • +23:00:20.898		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2017iwe • 13:33:12.093 • -16:07:05.938		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019xvv • 14:24:19.230 • -16:22:19.315		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020rvf • 03:54:15.942 • +15:55:42.899		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2021bwq • 11:28:17.079 • -12:53:17.606		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0

[Target • 2018hcp • 08:22:57.667 • +27:42:10.354](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2019nyk • 00:15:15.198 • -08:11:21.844](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2021gdz • 11:53:19.030 • +20:06:30.902](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2019ply • 02:59:46.978 • +15:17:48.653](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2020yyz • 02:34:00.175 • +20:58:26.634](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2019spk • 10:02:54.390 • -06:12:26.042](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2019xvo • 03:58:22.726 • +06:42:55.476](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2018hwm • 08:21:28.182 • +03:09:52.427](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2020aeje • 00:17:11.964 • -07:37:37.326](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2020nap • 12:07:08.824 • +16:55:16.230](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2018jkb • 02:21:09.805 • -04:04:23.657](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s

Total Tel. Time: 3600s		Signal/Noise: 0.0
Target • 2021gno • 12:12:10.298 • +13:14:57.052		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019dke • 12:53:38.211 • -26:17:31.391		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020tsc • 08:07:53.306 • -18:41:58.362		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018iqx • 22:27:53.594 • +19:27:38.484		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2017jcu • 03:22:46.967 • +00:08:53.585		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2017kam • 13:42:43.917 • -24:20:06.302		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018hi • 08:53:40.573 • -25:03:31.961		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020aaxf • 08:20:41.967 • -01:24:52.348		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020uzq • 23:34:00.196 • -11:46:58.984		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020acbm • 02:40:17.795 • +02:25:37.542		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018bjo • 12:51:19.856 • +21:13:49.040		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s	WFM-NOAO Instrument Mode: WFM-NOAO-N	Observation Integration Time: 0 s

Repeat: 4 x Total Tel. Time: 3600s	Telescope Overheads: 360 s	Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019zc • 00:29:29.399 • -13:28:42.733		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019hgu • 00:50:24.894 • -12:05:42.673		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020zoq • 10:36:21.592 • +13:43:07.205		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2021bjk • 02:01:52.221 • -10:27:48.906		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018kfn • 16:02:42.480 • +15:14:11.263		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020abtf • 07:59:34.479 • +15:18:00.025		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2021iy • 11:18:31.673 • -06:16:40.501		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018afm • 07:59:25.015 • +16:25:34.619		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2017hca • 08:49:41.068 • -08:05:31.218		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018hle • 03:39:28.109 • -13:07:02.384		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019tts • 08:02:49.927 • +15:48:05.537		Tel. Time: 01h00m
OS 1	WFM-NOAO	Observation

Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019pjs • 18:04:40.344 • +21:38:04.272		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2021aek • 11:59:48.027 • -21:23:12.905		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020xqj • 12:03:49.533 • +29:43:01.790		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020vzv • 21:13:24.762 • +17:40:29.374		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020htt • 10:31:38.369 • -17:51:08.114		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020jjj • 14:31:19.635 • -25:39:31.111		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020bij • 10:55:11.698 • -26:08:24.518		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018bui • 14:20:43.446 • +13:41:43.343		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2017ilo • 03:31:20.633 • +04:07:54.584		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018lcd • 01:51:10.660 • -03:29:24.853		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018cho • 00:13:26.503 • +17:29:19.543		Tel. Time: 01h00m

OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020szs • 21:34:08.994 • +08:46:46.366		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019env • 09:09:35.638 • +29:58:14.534		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019hnl • 23:43:10.247 • -02:56:58.873		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020adhh • 04:43:46.158 • +28:58:10.564		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020aer • 04:48:35.429 • -04:49:19.625		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019rrs • 09:37:08.616 • +09:27:55.530		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018aex • 11:00:45.374 • +22:17:14.741		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018cqi • 13:57:40.915 • +17:30:29.142		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020able • 09:26:02.269 • +24:31:18.152		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2021duk • 05:23:17.173 • -14:01:47.262		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0

[Target • 2018gle • 04:25:03.013 • -14:36:47.804](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2019wbd • 23:20:35.175 • -00:52:51.074](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2020xcu • 08:36:19.903 • -11:01:08.119](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2020avf • 11:02:59.468 • -16:17:31.682](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2019gql • 00:51:40.856 • -13:19:05.704](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2020adow • 08:33:42.263 • +27:42:43.528](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2019eez • 06:16:22.222 • -21:21:35.712](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2020uas • 22:59:38.659 • +20:08:35.686](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2020hox • 19:42:26.160 • -15:52:44.018](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2018bif • 11:45:44.824 • -11:47:17.041](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s
Total Tel. Time: 3600s		Signal/Noise: 0.0

[Target • 2021ejn • 10:37:14.439 • +04:33:07.682](#)

Tel. Time: 01h00m

OS 1	WFM-NOAO	Observation
Tel. Time: 900 s	Instrument Mode: WFM-NOAO-N	Integration Time: 0 s
Repeat: 4 x	Telescope Overheads: 360 s	Instrument Overheads: 0 s

Total Tel. Time: 3600s		Signal/Noise: 0.0
Target • 2020abf • 09:18:05.927 • -15:48:44.708		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019osl • 04:38:54.326 • +18:50:10.201		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019bzd • 14:47:32.040 • -19:45:57.629		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019tba • 04:55:09.845 • -16:09:01.451		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018rz • 14:27:11.437 • -02:16:11.060		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020kxg • 13:20:00.505 • -16:39:40.295		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020acat • 11:50:41.228 • -10:13:38.892		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2018jsn • 11:59:57.007 • -00:33:01.728		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019lsj • 19:36:59.137 • -11:57:13.655		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2019qba • 15:37:44.726 • +22:25:36.833		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2017gww • 00:03:49.218 • +07:28:00.289		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s	WFM-NOAO Instrument Mode: WFM-NOAO-N	Observation Integration Time: 0 s

Repeat: 4 x Total Tel. Time: 3600s	Telescope Overheads: 360 s	Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020ltq • 21:44:34.458 • -24:36:10.127		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0
Target • 2020ekk • 16:44:45.150 • +22:30:47.894		Tel. Time: 01h00m
OS 1 Tel. Time: 900 s Repeat: 4 x Total Tel. Time: 3600s	WFM-NOAO Instrument Mode: WFM-NOAO-N Telescope Overheads: 360 s	Observation Integration Time: 0 s Instrument Overheads: 0 s Signal/Noise: 0.0